Application Session

Bit by Bit Coding - Term 2 | Taster 4

Key Concepts

- Combine variables, control flow, and loops to build small programs.
- A clear problem-solving workflow keeps code manageable.
- Break big problems into small, testable steps.
- Debug systematically: read the error, find the bug, fix, re-test.

Syntax Reference

Combining Skills - a Simple Quiz

```
score = 0
for i in range(3):
    answer = input("Answer: ")
    if answer == "python":
        score += 1
print(f"You scored {score}/3")
```

Problem-Solving Workflow

- 1. Understand - read the problem; identify inputs and outputs.
- 2. Plan - break it into steps; write them as comments.
- 3. Code - implement one small step at a time.
- 4. Test - run with sample inputs; check edge cases.
- 5. Refine - tidy up and improve readability.

Common Patterns

- Use comments (#) to outline your steps before coding.
- Give variables clear names (total_score, not x).
- Test small pieces in isolation, then join them.

Tips & Common Mistakes

- Read error messages fully - they name the line and the problem.
- Print variables to see their values while debugging.
- Build a little, test a little - don't write everything then test.
- Indentation errors usually mean a missing or extra space or colon.